

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – SIH26070

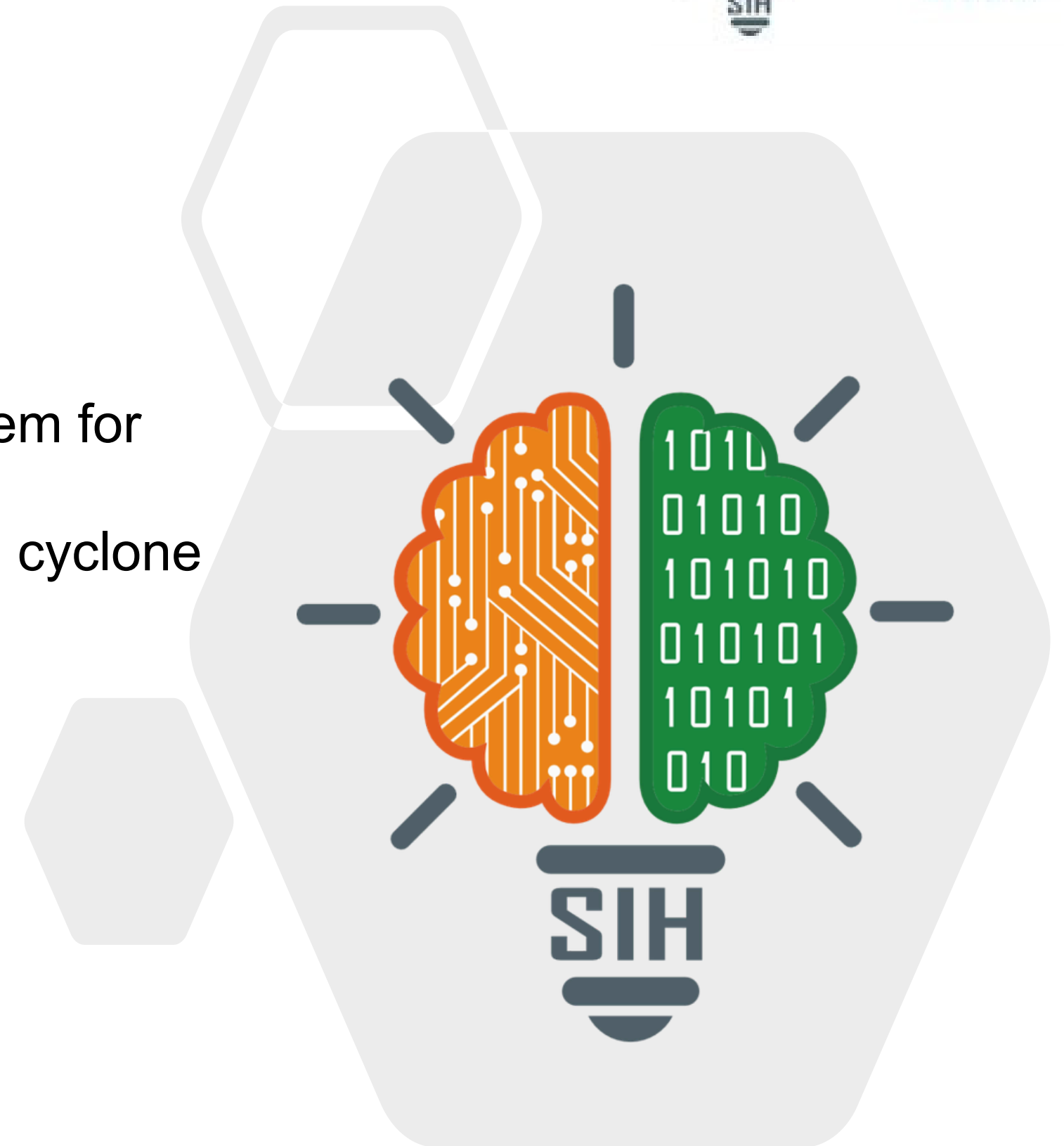
Problem Statement Title – To develop an AI/ML based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data.

Theme – Disaster Management

PS Category – Software

Team ID –

Team Name (Registered on portal) –Tech Wizards



N.E.T.R.A.

Neural Engine for Tropical Recognition and Analysis

- **Detailed Explanation:**

- **Dual-Stream Architecture:** We engineered a two-stream neural network that simultaneously processes Brightness Temperature (IR) and RAW Visible satellite imagery to estimate cyclone wind speeds.
- **Feature Extraction:** Utilizes a **Swin Transformer (Tiny)** to capture global thermal gradients from IR data, and **EfficientNet-B0** to extract localized cloud structures from visible imagery.

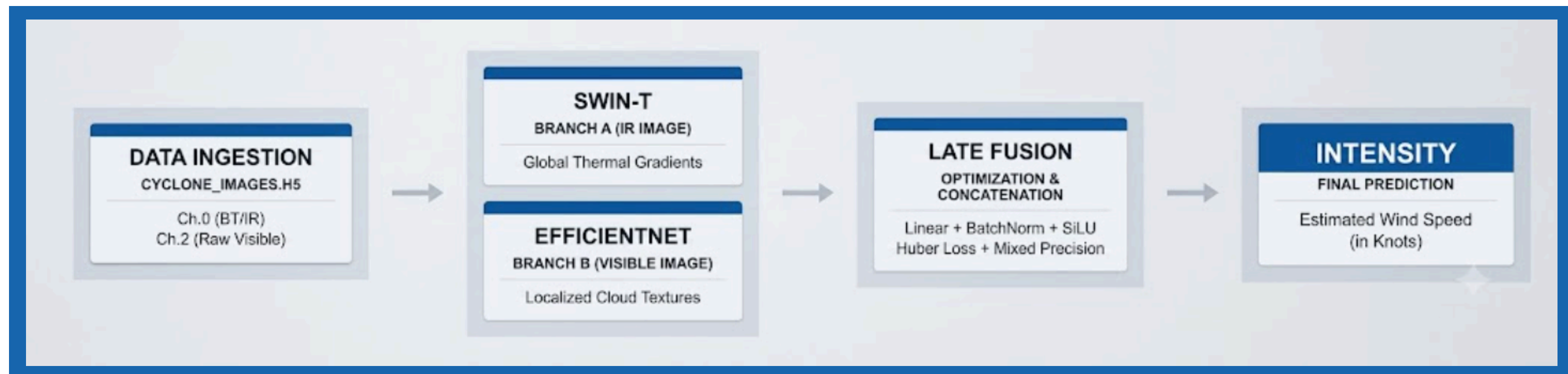
- **How it addresses the problem:**

- Replaces subjective manual chart analysis with an objective, automated intensity estimation pipeline that outputs exact continuous wind speed (in knots).

- **Innovation & Uniqueness:**

- **Late Fusion MLP:** We strip the classification heads off both models, concatenate their latent representations, and pass them through a deep Multi-Layer Perceptron (MLP) with SiLU activations.
- **Test-Time Augmentation (TTA):** The model employs dynamic horizontal flipping during live inference, averaging predictions to drastically reduce error margins (MAE).

- **Technologies to be used (e.g. programming languages, frameworks, hardware):**
 - **Languages & Libraries:** Python, PyTorch (torch.amp Mixed Precision), h5py, NumPy, OpenCV.
 - **Models & Optimizers:** Swin-T, EfficientNet-B0, AdamW, CosineAnnealingLR.
 - **Hardware:** CloudGPU(FP16 optimized) processing large HDF5 multidimensional arrays.
- **Methodology and process for implementation (Flow Charts/Images/working prototype):**



Feasibility

- **Efficient Data Pipeline:** Utilizes an optimized h5py DataLoader with persistent handles and multi-processing (num_workers=4), allowing massive satellite datasets to be trained and evaluated locally.
- **Hardware Efficiency:** Utilizes torch.autocast and GradScaler for mixed-precision (FP16) training, halving VRAM usage and making deployment on standard cloud GPUs highly feasible.

Potential Risks & Challenges	Overcoming these Risks
Outlier Sensitivity: Cyclones with extreme wind speeds (outliers/super cyclones) severely skew standard Mean Squared Error (MSE) loss functions, crashing model accuracy.	Huber Loss Implementation: We replaced standard loss with HuberLoss(delta=1.0). It acts as an L1 loss for large outlier errors and an L2 loss for small errors, providing extreme mathematical resilience to anomalous storms.

Potential impact on the target audience

- **Primary Users:** Empowers the India Meteorological Department (IMD) and National Disaster Management Authority (NDMA) with high-confidence forecasting.
- **Secondary Users:** Assists maritime shipping, Coast Guard, and vulnerable coastal zone populations across the subcontinent.

Benefits of the Solution		
Mission Zero Casualty Delivers near-instantaneous, objective cyclone intensity estimations (in knots) to disaster agencies, bypassing human cognitive fatigue.	High Mathematical Accuracy The combination of Test-Time Augmentation (TTA) and dual-sensor fusion drastically lowers the Mean Absolute Error (MAE) and RMSE.	Scalable Architecture The lightweight .pt (PyTorch) model state dictionary can be deployed instantly for live inference without retraining.

Details / Links of the reference and research work:

○ Datasets & Repositories:

- KaggleHigh-ResolutionCyclone Archive: HDF5 (.h5) geospatial matrix structures (Infrared Thermal & Raw Visible) & normalized label arrays. Link → <https://www.kaggle.com/datasets/vaukaofworlds/thecycloneimagedataset>

○ Foundational Academic Research:

- Liu, Z., et al. "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows." .
- Tan, M., & Le, Q. V. "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks."
- Huber, P. J. "Robust Estimation of a Location Parameter." *Annals of Mathematical Statistics*.

○ Software Frameworks:

- **PyTorch**: Framework for Mixed Precision (AMP) and Deep Learning.
- **h5py**: Pythonic interface to the HDF5 binary data format.